Input: $x \in \{0,1\}^n$

Output: $a \in \{0,1\}$

$v \leftarrow$ root of T

$p \leftarrow ?^n$

while v is not a leaf **do**

  $D^{v,p} \leftarrow \{j \in [n] : p_j = ? \text{ and } \mathrm{rank}(Q^{\preceq v}_{S(p,\star)+j}) = \mathrm{rank}(Q^{\preceq v}_{S(p,\star)}) + 1\}$

  if $D^{v,p} = \emptyset$ **then**

    Infer $b^v \leftarrow \langle x, Q^v \rangle$ from the fact that $(Q^{\prec v})^T x = b^{\prec v}$ and $x_j = 0$ for all $p_j = 0$

  else

    for $j \in D^{v,p}$ **do**

      Query x_j

      Sample $\boldsymbol{\eta} \sim \mathsf{Ber}(\delta_j/(2-\delta_j))$

      if $x_j = 1$ or $\boldsymbol{\eta} = 1$ **then**

        $p_j \leftarrow \star$

        break

      end if

      $p_j \leftarrow 0$

    end for

    Query $b^v \leftarrow \langle x, Q^v \rangle$

  end if

  Move $v \leftarrow \mathrm{child}(v, b^v)$

end while

return $L(v)$